What Does Swiss German Cost a Language Model? Separating Dialect from Spelling in LLM Normalization

Erik Avtandilyan and Simone Martinecz and Kelvin Myers
University of Zurich
Essentials in Text and Speech Processing, HS 2026
https://github.com/Eroavt/gsw-eval

Abstract

Swiss German is spoken by about five million people but has no standard spelling, so every writer spells it their own way. Large language models (LLMs) understand it worse than Standard German and need more tokens for it. We ask how much of this cost comes from the dialect and how much from the spelling. We build GSW-EVAL, an open, reproducible benchmark that feeds the same 500 sentences to a model as Standard German, as careful Swiss German in eight dialects, and as texting-style and character-noised Swiss German. On SwissDial, the dialect costs the 8B models 20 to 24 chrF points but Claude Sonnet 5.5 only 8.6; texting spelling adds just 1 to 5 points. At equal size, the Swiss model Apertus beats Llama 3.1 by 6.4 points, and the three times larger Gemma 3 27B is only 1.1 points ahead of Apertus; Claude is 11.5 points ahead of Apertus on dialect input while the two tie on Standard German. A second test set, Swiss German NTREX, confirms that spelling costs little. Token counts tell the same story: Swiss German needs 16 to 30% more tokens than Standard German, while texting spelling changes the count by about 1%. Per-dialect difficulty follows how close each written dialect is to Standard German ($r = 0.93$). **[TODO: one sentence on the fresh set and the human ratings once they are in.]**

1 Introduction

Written Swiss German is everywhere in Switzerland: in messages, social media and comments. It is also a textbook case of a low-resource variety inside a high-resource language: Standard German is one of the best-served languages in NLP, its dialects are not (Joshi et al., 2020). Two things make Swiss German hard for a language model. First, the *dialect*: the eight major dialects differ in vocabulary, sound and grammar. Second, the *spelling*: there is no orthography, so the same writer may spell a word differently from one message to the next.

Users pay for both, in quality and in money, since API prices are per token and non-standard text splits into more tokens (Petrov et al., 2023; Ahia et al., 2023). But the two causes need different fixes. A dialect problem calls for more dialect data; a spelling problem could be solved with normalization or with robust training. Existing work measures one or the other. We measure both on the same sentences.

Contributions.

1. GSW-EVAL, an open benchmark that runs any LLM on written Swiss German and splits the quality loss into a *dialect cost* and a *spelling cost*, with confidence intervals.
2. A comparison of a Swiss and a non-Swiss model of the same size, a larger open model and a commercial model, showing that Swiss training data is worth about 6 chrF points at 8B scale and brings an 8B model within 1.1 points of one three times its size, but that a frontier model loses less than half as much to the dialect.
3. A token-tax analysis of nine tokenizers, showing that the extra tokens come from the dialect, not from the spelling.
4. A small fresh test set in which each native writer translates the same sentences twice, carefully and in texting style. **[TODO: size once collected]**

2 Related Work

[TODO: Kelvin: rewrite this section in your own words, 0.5 column.]

Swiss German NLP. Earlier work normalizes Swiss German transcripts to a common spelling with character-level machine translation (Scherrer and Ljubešić, 2016) on the ArchiMob corpus

(Samardžić et al., 2016). SwissBERT (Vamvas et al., 2023) adapts an encoder to Swiss national languages, and Vamvas et al. (2024) adapt encoders to written Swiss German. SwissDial (Dogan-Schönberger et al., 2021) provides the same sentences in eight dialects with Standard German references, which makes a controlled comparison across dialects possible. Collecting dialect data remains the bottleneck (Sun et al., 2025).

Dialect and noise robustness. Aepli et al. (2023) show that standard MT metrics are unreliable when the output is Swiss German. Our outputs are Standard German, where chrF is on firmer ground, but we still check it against human ratings. DialectBench (Faisal et al., 2024) shows large gaps between standard and non-standard varieties across many tasks. Synthetic character noise breaks neural translation (Belinkov and Bisk, 2018), and injecting such noise during training helps transfer between closely related languages (Aepli and Sennrich, 2022). We use the same noise for evaluation and compare it with rule-based texting spelling.

Tokenization cost. The same content needs very different numbers of tokens in different languages (Rust et al., 2021; Petrov et al., 2023), which makes commercial models more expensive for some users (Ahia et al., 2023). We apply this lens *within* one language: Standard German versus its dialects.

3 Data

SwissDial. SwissDial v1.1 (Dogan-Schönberger et al., 2021) contains 11,213 Standard German sentences with transcriptions in up to eight dialects (AG, BE, BS, GR, LU, SG, VS, ZH), one writer per dialect. We keep the 2,529 sentences available in all eight dialects and drop 232 marked as code-switching, leaving 2,297. We draw a fixed split of 50 development and 500 test sentences (seed 13), giving 4,000 Swiss German test inputs per condition. The Standard German sentence is always the reference.

Swiss German NTREX. To check that our findings are not specific to SwissDial, we repeat the evaluation on a second, independent test set. Aepli et al. (2023) had two professional translators translate the 1,997 English news sentences of NTREX-128 into Bernese and Zurich German; NTREX-128 itself provides a Standard German translation of the same sentences, which serves as the reference. 1,850 sentences can be aligned in both dialects, and we fix 500 of them as a second test split. Because both the Swiss and the Standard German versions are translated from English, the reference is looser than in SwissDial, so absolute scores are lower; the comparisons between models and conditions remain valid.

Fresh set. **[TODO: Simone: who wrote it (number of writers, dialects), how long it took.]** Each writer translated the same 30 Standard German sentences (20 everyday chat, 10 news-like) twice: once carefully and once "the way you would text". Because the same person writes both versions, the difference between them is spelling only. The set was written after the models were trained, so it cannot be in their training data. It is released with the code (CC BY 4.0).

4 Method

Conditions. Every test sentence is given to each model in four forms: (1) *Standard German input*: the reference itself; (2) *clean*: the SwissDial transcription; (3) *texting*: the clean text rewritten by 24 texting rules (lower case, dropped punctuation, collapsed double vowels, short forms such as *vlt* for *vilicht*), each firing with a probability, **[TODO: Simone: confirm the rules are realistic]**; and (4) *character noise*: 10% of words get one random letter insertion, deletion or substitution, following Aepli and Sennrich (2022). All noise is seeded per sentence.

Baselines. The *identity* baseline copies the Swiss German input unchanged; it is the lower bound every model must beat. The *Standard German input* condition is the upper bound: the best a model can do when there is no dialect and no spelling problem.

Dialect and spelling cost. We define the *dialect cost* as chrF on Standard German input minus chrF on clean Swiss German, and the *spelling cost* as chrF on clean minus chrF on texting input. Both are computed on the same sentences, so content is held constant.

Models. Apertus 8B Instruct (Hernández-Cano et al., 2025), trained with a strong Swiss and multilingual focus; Llama 3.1 8B Instruct (Grattafiori et al., 2024), the same size without a Swiss focus; and Gemma 3 27B Instruct (Gemma Team, 2025), a larger open model without a Swiss focus. All three run locally with 8-bit weights and greedy decoding via MLX. Claude Sonnet 5.5 runs through its API with the lowest thinking setting, as a commercial reference. All models receive the same

German prompt asking for the Standard German sentence only.

Metric and statistics. We score with chrF (Popović, 2015) as implemented in sacreBLEU (Post, 2018); character n-grams suit a task where most errors are partial word forms. All intervals are 95% bootstrap intervals over sentences (Koehn, 2004); model differences use a paired bootstrap, and the costs use a cluster bootstrap that resamples whole sentences with all their dialects. SwissDial writes numbers as words (*zweitausendelf*) while models mostly write digits (*2011*); we therefore score against both versions of the reference, which SwissDial provides, and keep the better match (multi-reference chrF). We also remove answer preambles such as "Der Satz lautet auf Standarddeutsch:".

Token tax. For a tokenizer, the token tax is the number of tokens of the Swiss German sentences divided by the tokens of the same sentences in Standard German, summed over all 2,297 parallel sentences and eight dialects. Because the texting rules drop punctuation and punctuation marks are tokens, we compare all conditions with punctuation removed.

Human evaluation. **[TODO: Kelvin + Simone: two native speakers rated 96 outputs (48 inputs x 2 models, blind, random order) for adequacy on a 1–5 scale; report kappa here.]**

5 Results

The dialect costs far more than the spelling. Table 1 shows that moving from Standard German to clean Swiss German costs Apertus 19.9 and Llama 23.8 chrF points, but Claude only 8.6. Texting spelling costs a further 3.5, 5.0 and 1.1 points, character noise 2.9, 3.0 and 1.4. All models stay far above the identity baseline (46.9), so they translate rather than copy.

Swiss data helps; size helps less; a frontier model helps most. Apertus beats Llama in every condition (paired bootstrap, $p < 0.01$): by 2.5 points on Standard German input, 6.4 on clean dialect and 7.9 on texting input, so the advantage grows where the input is least standard. Gemma 3 27B, three times larger but without a Swiss focus, is only 1.1 points ahead of Apertus on clean dialect and 2.0 on texting input. Claude is 11.5 points ahead of Apertus on clean dialect and 13.9 on texting input, but the two tie on Standard German input (+0.2, interval −1.1 to +1.5): Claude's lead

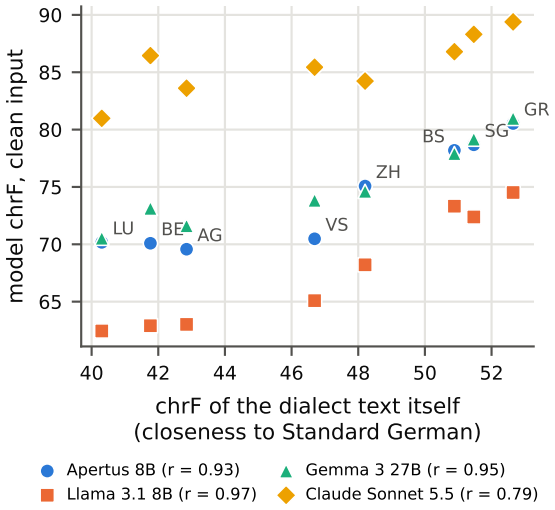


Figure 1: Per-dialect chrF of each model (clean input) against the chrF of the untranslated dialect text, a measure of how close its spelling is to Standard German.

is entirely about the dialect.

Even Standard German is not copied. No model reaches 100 on Standard German input: Apertus returns the sentence unchanged 64% of the time, Claude 59%, Llama 45% and Gemma only 23%; the rest are paraphrases such as "Jemand vermiest mir hier die Arbeit" for "Irgendjemand pfuscht mir hier ins Handwerk". This is why the upper bound matters: part of every model's gap to 100 is paraphrase that chrF counts as error, not misunderstanding. It also means a small dialect cost alone is not good news. Gemma's dialect cost (9.4) is almost as small as Claude's (8.6), but mainly because Gemma rewrites even Standard German (84.7 against Claude's 94.2), not because it reads dialect well. The dialect cost has to be read together with the upper bound. **[TODO: the human ratings tell how big the paraphrase part is.]**

Hard dialects are the ones written farthest from Standard German. Scores range from 69.6 (AG) to 80.5 (GR) for Apertus, and the ranking is almost the same for Llama and Gemma (Figure 1). All three open models correlate strongly with the identity score of each dialect ($r = 0.93$, 0.97 and 0.95): models do worst where the written form is farthest from Standard German. Claude depends less on it ($r = 0.79$, range 81.0 to 89.4). Since SwissDial has one writer per dialect, this mixes the dialect with the writer's personal spelling. The fresh set separates the two. **[TODO: fresh-set result.]**

Token tax comes from the dialect, not the

Model	Std. German	Clean	Texting	Char. noise	Dialect cost	Spelling cost
Identity (lower bound)	100.0 ±0.0	46.9 ±0.4	40.0 ±0.3	45.3 ±0.4	53.1 ±1.0	6.8 ±0.4
Apertus 8B	94.0 ±1.0	74.1 ±0.5	70.6 ±0.6	71.3 ±0.5	19.9 ±1.2	3.5 ±0.5
Llama 3.1 8B	91.5 ±1.0	67.7 ±0.6	62.7 ±0.6	64.8 ±0.6	23.8 ±1.2	5.0 ±0.6
Gemma 3 27B	84.7 ±1.1	75.2 ±0.5	72.6 ±0.5	73.3 ±0.5	9.4 ±1.2	2.6 ±0.5
Claude Sonnet 5.5	94.2 ±1.0	85.6 ±0.4	84.5 ±0.4	84.2 ±0.4	8.6 ±1.1	1.1 ±0.3

Table 1: chrF on the 500 test sentences (±: half-width of the 95% bootstrap interval). Swiss German columns pool 4,000 inputs over eight dialects. Dialect cost = Standard German input minus clean; spelling cost = clean minus texting.

Tokenizer	Tok./word	Tax	Range	Texting
Apertus 8B	1.64	1.30	1.23–1.40	+0.3%
Llama 3.1 8B	1.93	1.17	1.12–1.25	-1.3%
Gemma 3	1.62	1.25	1.20–1.35	+0.0%
Qwen 3	1.92	1.18	1.12–1.26	-1.0%
Mistral 7B v0.3	2.10	1.16	1.11–1.23	-0.7%
OpenAI o200k	1.60	1.27	1.23–1.37	-0.4%
XLM-R	1.63	1.30	1.25–1.39	+0.8%
SwissBERT	1.43	1.59	1.48–1.70	+7.1%

Table 2: Token tax per tokenizer. Tok./word: tokens per word of Standard German. Tax: tokens for Swiss German divided by tokens for the same Standard German sentences (punctuation removed, eight dialects pooled). Range: cheapest to most expensive dialect. Texting: change of the tax caused by texting spelling.

Model	Std. DE	Clean	Texting	Noise	Spelling
Identity	100.0	38.6	33.4	37.5	5.2
Apertus 8B	97.0	56.6	54.9	55.4	1.8
Llama 3.1 8B	93.6	53.8	51.5	52.5	2.3
Gemma 3 27B	84.3	57.5	56.2	56.5	1.3
Claude Sonnet 5.5	96.0	59.9	59.5	59.5	0.4

Table 3: chrF on the 500 Swiss German NTREX test sentences (Bernese and Zurich German pooled; intervals ±0.2 to 1.1). Noise: character noise. Spelling: spelling cost (clean minus texting).

spelling. Swiss German needs 16% (Mistral) to 30% (Apertus, XLM-R) more tokens than the same content in Standard German (Table 2). Texting spelling changes this by about 1% for every LLM tokenizer: it creates rarer word forms but also shortens words, and the two effects cancel. Bernese is the most expensive dialect and Zurich German the cheapest for all LLM tokenizers. Apertus pays the highest *relative* tax only because its tokenizer is very efficient on Standard German (1.64 tokens per word against 1.93 for Llama); in absolute terms it needs fewer tokens for Swiss German than Llama. Claude's tokenizer behaves differently: it needs about 3.3 tokens per word for Standard and Swiss German alike, so Swiss German costs no extra there, but every German sentence is expensive (test split, counted with Anthropic's token-counting endpoint).

The findings hold on a second test set. On Swiss German NTREX (Table 3), texting spelling again costs little: 0.4 (Claude) to 2.3 points (Llama). The model ranking is the same as on SwissDial (Claude, Gemma, Apertus, Llama on clean input; each neighbouring pair differs with $p < 0.001$), and Apertus again beats Llama on every condition (2.9 points on clean input), although the gaps are smaller. We do not interpret the NTREX dialect cost, because its Swiss and Standard German versions are independent translations from English, so part of the gap is wording, not dialect.

Robustness. Across three prompt wordings (100 test sentences), clean-dialect chrF moves by 0.2 points for Claude, 2.2 for Llama and 3.1 for Apertus. Apertus stays ahead of Llama under every wording, and the spelling cost stays between 3.1 and 5.3 points for both 8B models, far below their dialect cost. Full-precision (bf16) weights change the 8-bit scores of Apertus and Llama by at most 0.5 points.

Human ratings. **[TODO: kappa, Spearman chrF vs rating, and whether chrF picks the same better model as the raters.]**

Fresh set. **[TODO: real vs synthetic texting cost, per writer.]**

6 Discussion and Limitations

[TODO: Kelvin + Erik: write after all results are in. Points to cover:] (1) SwissDial transcriptions are free translations by one writer per dialect, so dialect and writer are confounded; the fresh set addresses this only for **[TODO: N]** writers. (2) The texting rules are hand-written and applied with

fixed probabilities; real texting is richer (abbreviations, emoji, code-switching). (3) chrF punishes valid paraphrases; the human ratings estimate how much. (4) We test one prompt per model in the main run and 8-bit weights; the robustness checks cover both on a subset. (5) SwissDial has been public since 2021 and may be in the models' training data; the fresh set cannot be. (6) A small dialect cost can come from a weak upper bound: a model that paraphrases Standard German (Gemma) looks robust to dialect. The cost must be read with the upper bound and copy rate.

7 Conclusion

What does Swiss German cost a language model? Mostly the dialect, not the spelling. Moving from Standard German to careful Swiss German costs 8B models 20 to 24 chrF points and needs 16 to 30% more tokens, while texting spelling adds only 1 to 5 points and about 1% of tokens. Swiss training data recovers about 6 points at equal size, nearly as much as a model three times larger, and a frontier model cuts the dialect cost to 8.6. Both findings replicate on a second test set. For users, this means that cleaning up spelling buys little; choosing a model that knows the dialect buys a lot. **[TODO: one sentence on the fresh set once it is in.]**

Code and Data

Code, outputs of every model, and the fresh set: `https://github.com/Eroavt/gsw-eval`. Every number in this report can be reproduced with one command, `scripts/reproduce.sh`, which needs neither a GPU nor an API key because all model outputs are cached.

Use of AI Assistants

We used an AI coding assistant (Claude Code) to write and test the code and to draft parts of this report. All design decisions, the data collection and the final text are our own, and we checked every number against the code's output. **[TODO: adapt to UZH rules.]**

References

Noëmi Aepli, Chantal Amrhein, Florian Schottmann, and Rico Sennrich. 2023. A benchmark for evaluating machine translation metrics on dialects without standard orthography. In *Proceedings of the Eighth Conference on Machine Translation*, pages 1045–1065.

Noëmi Aepli and Rico Sennrich. 2022. Improving zero-shot cross-lingual transfer between closely related languages by injecting character-level noise. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 4074–4083.

Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David Mortensen, Noah Smith, and Yulia Tsvetkov. 2023. Do all languages cost the same? tokenization in the era of commercial language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9904–9923.

Yonatan Belinkov and Yonatan Bisk. 2018. Synthetic and natural noise both break neural machine translation. In *International Conference on Learning Representations (ICLR)*.

Pelin Dogan-Schönberger, Julian Mäder, and Thomas Hofmann. 2021. SwissDial: Parallel multidialectal corpus of spoken Swiss German. *arXiv preprint arXiv:2103.11401*.

Fahim Faisal, Orevaoghene Ahia, Aarohi Srivastava, Kabir Ahuja, David Chiang, Yulia Tsvetkov, and Antonios Anastasopoulos. 2024. DIALECTBENCH: An NLP benchmark for dialects, varieties, and closely-related languages. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 14412–14454.

Gemma Team. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*.

Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Alejandro Hernández-Cano et al. 2025. Apertus: Democratizing open and compliant LLMs for global language environments. *arXiv preprint arXiv:2509.14233*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293.

Philipp Koehn. 2004. Statistical significance tests for machine translation evaluation. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395.

Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. 2023. Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Maja Popović. 2015. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395.

Matt Post. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191.

Phillip Rust, Jonas Pfeiffer, Ivan Vulić, Sebastian Ruder, and Iryna Gurevych. 2021. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pages 3118–3135.

Tanja Samardžić, Yves Scherrer, and Elvira Glaser. 2016. ArchiMob – a corpus of spoken Swiss German. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*.

Yves Scherrer and Nikola Ljubešić. 2016. Automatic normalisation of the Swiss German ArchiMob corpus using character-level machine translation. In *Proceedings of the 13th Conference on Natural Language Processing (KONVENS 2016)*, pages 248–255.

J. Sun, R. Sevastjanova, S. Ahmadi, R. Sennrich, and M. El-Assady. 2025. Dia-Lingle: A gamified interface for dialectal data collection. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (System Demonstrations)*, pages 148–158.

Jannis Vamvas, Noëmi Aepli, and Rico Sennrich. 2024. Modular adaptation of multilingual encoders to written Swiss German dialect. In *Proceedings of the 1st Workshop on Modular and Open Multilingual NLP (MOOMIN 2024)*, pages 16–23.

Jannis Vamvas, Johannes Graën, and Rico Sennrich. 2023. SwissBERT: The multilingual language model for Switzerland. In *Proceedings of the 8th Swiss Text Analytics Conference (SwissText 2023)*, pages 54–69.